

ТРИГОНОМЕТРИЯ

Тригонометрични функции на остър ъгъл

Определения

$$\left\{ \begin{array}{l} \sin \alpha = \frac{a}{c} \\ \cos \alpha = \frac{b}{c} \\ \operatorname{tg} \alpha = \frac{a}{b} \\ \operatorname{cotg} \alpha = \frac{b}{a} \end{array} \right. \quad \left\{ \begin{array}{l} \sin \beta = \frac{b}{c} \\ \cos \beta = \frac{a}{c} \\ \operatorname{tg} \beta = \frac{b}{a} \\ \operatorname{cotg} \beta = \frac{a}{b} \end{array} \right.$$

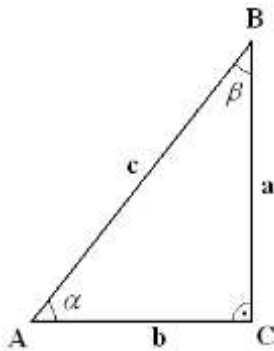


Таблица на знаците

Квадрант:	$\sin \alpha$	$\cos \alpha$	$\operatorname{tg} \alpha$	$\operatorname{cotg} \alpha$
I	+	+	+	+
II	+	-	-	-
III	-	-	+	+
IV	-	+	-	-

Стойности на някои ъгли

α	0°	30°	45°	60°	90°	120°	150°	180°	270°	360°
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	-1	0
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	0	1
$\operatorname{tg} \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	Не е деф.	$-\sqrt{3}$	$-\frac{\sqrt{3}}{3}$	0	Не е деф.	0
$\operatorname{cotg} \alpha$	Не е деф.	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{\sqrt{3}}{3}$	$-\sqrt{3}$	Не е деф.	0	Не е деф.

Основни тъждества

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\operatorname{tg} \alpha \cdot \operatorname{cotg} \alpha = 1$$

$$\operatorname{tg} \alpha = \frac{\sin \alpha}{\cos \alpha}$$

$$\operatorname{cotg} \alpha = \frac{\cos \alpha}{\sin \alpha}$$

$$1 + \operatorname{tg}^2 \alpha = \frac{1}{\cos^2 \alpha}$$

$$1 + \operatorname{cotg}^2 \alpha = \frac{1}{\sin^2 \alpha}$$

Връзка между тригонометричните функции

	$\sin \alpha$	$\cos \alpha$	$\operatorname{tg} \alpha$	$\operatorname{cotg} \alpha$
$\sin \alpha =$	$\sin \alpha$	$\pm \sqrt{1 - \cos^2 \alpha}$	$\frac{\operatorname{tg} \alpha}{\pm \sqrt{1 + \operatorname{tg}^2 \alpha}}$	$\frac{1}{\pm \sqrt{1 + \operatorname{cotg}^2 \alpha}}$
$\cos \alpha =$	$\pm \sqrt{1 - \sin^2 \alpha}$	$\cos \alpha$	$\frac{1}{\pm \sqrt{1 + \operatorname{tg}^2 \alpha}}$	$\frac{\operatorname{cotg} \alpha}{\pm \sqrt{1 + \operatorname{cotg}^2 \alpha}}$
$\operatorname{tg} \alpha =$	$\frac{\sin \alpha}{\pm \sqrt{1 - \sin^2 \alpha}}$	$\frac{\pm \sqrt{1 - \cos^2 \alpha}}{\cos^2 \alpha}$	$\operatorname{tg} \alpha$	$\frac{1}{\operatorname{cotg} \alpha}$
$\operatorname{cotg} \alpha =$	$\frac{\pm \sqrt{1 - \sin^2 \alpha}}{\sin^2 \alpha}$	$\frac{\cos \alpha}{\pm \sqrt{1 - \cos^2 \alpha}}$	$\frac{1}{\operatorname{tg} \alpha}$	$\operatorname{cotg} \alpha$

	$-\alpha$	$90^\circ - \alpha$	$90^\circ + \alpha$	$180^\circ - \alpha$	$180^\circ + \alpha$	$270^\circ - \alpha$	$270^\circ + \alpha$	$k.360^\circ - \alpha$	$k.360^\circ + \alpha$
sin α	$-\sin \alpha$	$+\cos \alpha$	$+\cos \alpha$	$+\sin \alpha$	$-\sin \alpha$	$-\cos \alpha$	$-\cos \alpha$	$-\sin \alpha$	$+\sin \alpha$
cos α	$+\cos \alpha$	$+\sin \alpha$	$-\sin \alpha$	$-\cos \alpha$	$-\cos \alpha$	$-\sin \alpha$	$+\sin \alpha$	$+\cos \alpha$	$+\cos \alpha$
tg α	$-\operatorname{tg} \alpha$	$+\operatorname{cotg} \alpha$	$-\operatorname{cotg} \alpha$	$-\operatorname{tg} \alpha$	$+\operatorname{tg} \alpha$	$+\operatorname{cotg} \alpha$	$-\operatorname{cotg} \alpha$	$-\operatorname{tg} \alpha$	$+\operatorname{tg} \alpha$
cotg α	$-\operatorname{cotg} \alpha$	$+\operatorname{tg} \alpha$	$-\operatorname{tg} \alpha$	$-\operatorname{cotg} \alpha$	$+\operatorname{cotg} \alpha$	$+\operatorname{tg} \alpha$	$-\operatorname{tg} \alpha$	$-\operatorname{cotg} \alpha$	$+\operatorname{cotg} \alpha$

Сбор и разлика на 2 ъгъла

$$\sin(\alpha \pm \beta) = \sin \alpha \cdot \cos \beta \pm \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cdot \cos \beta \mp \sin \alpha \cdot \sin \beta$$

$$\operatorname{tg}(\alpha \pm \beta) = \frac{\operatorname{tg} \alpha \pm \operatorname{tg} \beta}{1 \mp \operatorname{tg} \alpha \cdot \operatorname{tg} \beta}$$

$$\operatorname{cotg}(\alpha \pm \beta) = \frac{\operatorname{cotg} \alpha \cdot \operatorname{cotg} \beta \mp 1}{\operatorname{cotg} \beta \pm \operatorname{cotg} \alpha}$$

Синусова теорема

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} = 2R$$

Косинусова теорема

$$a^2 = b^2 + c^2 - 2bc \cdot \cos \alpha$$

$$b^2 = a^2 + c^2 - 2ac \cdot \cos \beta$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos \gamma$$

Следствие 1

$$a = \frac{c \cdot \sin \alpha}{\sin(\alpha + \beta)}$$

$$b = \frac{c \cdot \sin \beta}{\sin(\alpha + \beta)}$$

$$c = \frac{a \cdot \sin(\alpha + \beta)}{\sin \alpha}$$

Следствие 2

$$\sin \alpha = \frac{a}{b} \sin \beta$$

$$\sin \beta = \frac{b}{a} \sin \alpha$$

$$\sin \gamma = \frac{c}{b} \sin \beta$$

Следствие 3

$$\cos \alpha = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos \beta = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos \gamma = \frac{a^2 + b^2 - c^2}{2ab}$$

Следствие 4

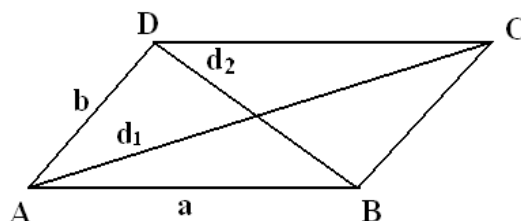
$$\gamma < 90^\circ \Leftrightarrow a^2 + b^2 > c^2$$

$$\gamma = 90^\circ \Leftrightarrow a^2 + b^2 = c^2$$

$$\gamma > 90^\circ \Leftrightarrow a^2 + b^2 < c^2$$

В успоредник

$$2(a^2 + b^2) = d_1^2 + d_2^2$$

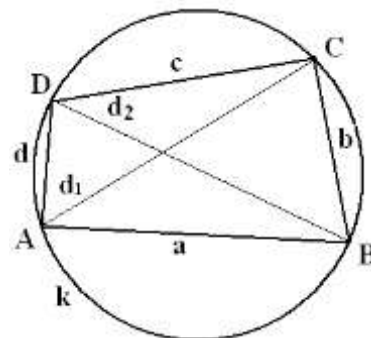


Произволен четириъгълник, вписан в $k(O, R)$

Условие: $AB + CD = AD + BC$

I Теорема на Птолемей: $d_1 d_2 = ac + bd$

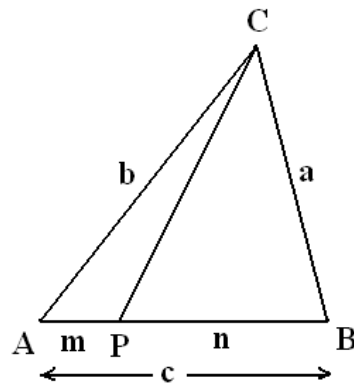
II Теорема на Птолемей: $\frac{d_1}{d_2} = \frac{ab + cd}{bc + ad}$



Теорема на Стюарт $CP^2 = \frac{ma^2 + nb^2}{c} - mn$

Ако Р е продължение на страната АВ =>

$$\begin{cases} CP^2 = \frac{-ma^2 + nb^2}{c} + mn, \text{ когато А е между Р и В;} \\ CP^2 = \frac{ma^2 - nb^2}{c} + mn, \text{ когато В е между А и Р.} \end{cases}$$



Формули за медианите в триъгълника

$$m_a = \frac{1}{2}\sqrt{2(b^2 + c^2) - a^2} \quad m_b = \frac{1}{2}\sqrt{2(a^2 + c^2) - b^2} \quad m_c = \frac{1}{2}\sqrt{2(a^2 + b^2) - c^2}$$

Формули за ъглополовящите в триъгълника

$$p = \frac{a+b+c}{2} \Rightarrow l_a = \frac{2}{b+c}\sqrt{bcp(p-a)} \quad l_b = \frac{2}{a+c}\sqrt{acp(p-b)} \quad l_c = \frac{2}{a+b}\sqrt{abp(p-c)}$$

Формула на Ойлер $d = \sqrt{R(R-2r)}$

Ако М е медицентър $\Rightarrow a^2 + b^2 + c^2 = 3(MA^2 + MB^2 + MC^2)$

Теорема на Лайбниц $PA^2 + PB^2 + PC^2 = MA^2 + MB^2 + MC^2 + 3 \cdot PM^2$
където Р е произволна точка (вътре, или извън триъгълника ABC).

Тангенсова Теорема $\frac{a+b}{a-b} = \frac{tg \frac{a+b}{2}}{tg \frac{a-b}{2}}$

Двойни и половинки ъгли

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha = 2 \cos^2 \alpha - 1$$

$$\cos \alpha \pm \sin \alpha = \sqrt{1 \pm \sin 2\alpha}$$

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$\sin \alpha = \frac{2tg \frac{\alpha}{2}}{1 + tg^2 \frac{\alpha}{2}}$$

$$\cos \alpha = \frac{1 - tg^2 \frac{\alpha}{2}}{1 + tg^2 \frac{\alpha}{2}}$$

Понижаване на степен

$$2 \sin^2 \alpha = 1 - \cos 2\alpha \quad 2 \cos^2 \alpha = 1 + \cos 2\alpha$$

Други

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2} = 2 \sin \frac{\alpha + \beta}{2} \sin \frac{\beta - \alpha}{2}$$